

materialsetup

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17:11

Contents

1	Routines used to set up the materials file	1
2	References	3
3	INDEX	4

1 Routines used to set up the materials file

\$Id: 17e67dc50d9b8c5ba07532bd44be3b97dab12f24 **\$**

The materials described in the file **materials_infile** identified in **degas2.in** are essentially just labels which will be used in conjunction with the plasma-material interactions to specify how test species interact with non-transparent surfaces.

Each material in **materials_infile** is described by two lines. The first is the full name (spaces allowed); the second is a corresponding symbol (no spaces). Note that any similarity between a material symbol and a species symbol is treated as coincidental. The motivations for being so general with materials while the descriptions of the species are so detailed is that first, the actual composition of the first wall will likely vary significantly from one device to another, even if they are nominally made of the substance (e.g., graphite). Second, the computational description of that surface has usually been generated by a code such as **TRIM**. The precise definition of the “material” used in that code is determined by the code’s input file and may not be the same that used by another code, or even another run of the same code. Hence, the names and symbols we use for our materials in DEGAS 2 need to be flexible to permit such variations to be distinguished.

For example,

```
carbon
C
```

```
iron
Fe
```

```
exit
Ex
```

```
mirror
mirror
```

```
molybdenum
mo
```

datasetup packages this information into DEGAS 2 form and writes it into the netCDF file corresponding to the symbolic name **materialsfile**.

```
"materialsetup.f" 1 ≡
  @m FILE 'materialsetup.web'
```

The unnamed module.

```
"materialsetup.f" 1.1 ≡
  〈Functions and Subroutines 1.2〉
```

Read in data from `materials_infile`.

⟨ Functions and Subroutines 1.2 ⟩ ≡

```

subroutine read_materials
  implicit none f77
  ma_common     // Common
  rf_common
  implicit none f90

  integer length, p, b, e     // Local
  character*LINELEN line
  character*FILELEN tempfile

  ⟨ Memory allocation interface 0 ⟩
  st_decls
  ma_num = 0
  materials_version = 'unknown'
  tempfile = filenames_array materials_infile
  assert(tempfile ≠ char_undef)
  open(unit = diskin, file = tempfile, form = 'formatted', status = 'old')
  assert(read_string(diskin, line, length))
  assert(length ≤ len(line))
  materials_version = line(: length)

loop1: continue
  if (¬read_string(diskin, line, length))
    goto eof     /* Add next material */
  ma_num = ma_num + 1
  var_realloca(materials_name)
  var_realloca(materials_sy)     /* First line contains material name */
  length = parse_string(line(: length))
  materials_name ma_num = line(: length)     /* Second line contains symbol */
  assert(read_string(diskin, line, length))
  length = parse_string(line(: length))     /* Extract species "symbol" from this line */
  p = 0
  assert(next_token(line, b, e, p))
  ma_sy(ma_num) = read_text(line(b: e))     /* And that should be it */
  assert(¬next_token(line, b, e, p))
  goto loop1
eof: continue
  close(unit = diskin)
  var_reallocb(materials_name)
  var_reallocb(materials_sy)

  return
end

```

See also section 1.3.

This code is used in section 1.1.

Write out data into netcdf file **materials.nc**

⟨Functions and Subroutines 1.2⟩ +=

```
subroutine nc_write_materials
  implicit_none_f77
  ma_common    // Common
  rf_common
  implicit_none_f90
  nc_decls    // Local
  integer fileid

  ma_ncdecl
  st_decls
  character*LINELEN description, program_version
  character*FILELEN tempfile    // local

  program_version = 'DEGAS_2_Git_commit:_%H$, _ref_names:_%d$'

  tempfile = filenames_array_materialsfile
  assert(tempfile ≠ char_undef)
  fileid = nccreate(tempfile, NC_CLOBBER, nc_stat)

  description = 'Data_for_materials_in_degas_2'
  call ncattputc(fileid, NC_GLOBAL, 'description', NC_CHAR, string_length(description),
    description, nc_stat)

  call ncattputc(fileid, NC_GLOBAL, 'data_version', NC_CHAR, string_length(materials_version),
    materials_version, nc_stat)

  call ncattputc(fileid, NC_GLOBAL, 'program_version', NC_CHAR,
    string_length(program_version), program_version, nc_stat)

  ma_ncdef(fileid)
  call ncendef(fileid, nc_stat)
  ma_ncwrite(fileid)
  call ncclose(fileid, nc_stat)

  return
end
```

2 References

3 INDEX

assert: 1.2, 1.3.

b: 1.2.

char_undef: 1.2, 1.3.

datasetup: 1.

description: 1.3.

diskin: 1.2.

e: 1.2.

eof: 1.2.

file: 1.2.

FILE: 1.

fileid: 1.3.

FILELEN: 1.2, 1.3.

filenames_array: 1.2, 1.3.

form: 1.2.

implicit_none_f77: 1.2, 1.3.

implicit_none_f90: 1.2, 1.3.

len: 1.2.

length: 1.2.

line: 1.2.

LINELEN: 1.2, 1.3.

loop1: 1.2.

ma_common: 1.2, 1.3.

ma_ncdecl: 1.3.

ma_ncdef: 1.3.

ma_ncwrite: 1.3.

ma_num: 1.2.

ma_sy: 1.2.

materials_infile: 1.2.

materials_name: 1.2.

materials_sy: 1.2.

materials_version: 1.2, 1.3.

materialsfile: 1.3.

NC_CHAR: 1.3.

NC_CLOBBER: 1.3.

nc_decls: 1.3.

NC_GLOBAL: 1.3.

nc_stat: 1.3.

nc_write_materials: 1.3.

ncattputc: 1.3.

ncclose: 1.3.

nccreate: 1.3.

ncendef: 1.3.

next_token: 1.2.

p: 1.2.

parse_string: 1.2.

program_version: 1.3.

read_materials: 1.2.

read_string: 1.2.

read_text: 1.2.

rf_common: 1.2, 1.3.

st_decls: 1.2, 1.3.

status: 1.2.

string_length: 1.3.

tempfile: 1.2, 1.3.

unit: 1.2.

var_realloca: 1.2.

var_reallocb: 1.2.

⟨ Functions and Subroutines 1.2, 1.3 ⟩ Used in section 1.1.

⟨ Memory allocation interface 0 ⟩ Used in section 1.2.

COMMAND LINE: "fweave -f -i! -W[-ybs15000 -ykw800 -ytw40000 -j -n/
/Users/dstotler/degas2/src/materialsetup.web".

WEB FILE: "/Users/dstotler/degas2/src/materialsetup.web".

CHANGE FILE: (none).

GLOBAL LANGUAGE: FORTRAN.